

JORDAN VICKERS

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EDUCATION

Rensselaer Polytechnic Institute, Troy, New York
PhD Physics

July 2025 - Present
GPA: 3.83/4.0

St. Olaf College, Northfield, Minnesota
BA Physics and Mathematics with Concentration in Statistics and Data Science

August 2021 - May 2025
GPA: 3.84/4.0

RESEARCH INTERESTS

- Open Quantum Systems: Modeling noise in quantum systems and applying optimal control to mitigate environmental noise and enhance gate performance.
- Quantum Full-Stack: Investigating the interplay between hardware-level qubit calibration and high-level software architectures.
- High-Energy Physics: Utilizing distributed computing and Python for large-scale data analysis

RESEARCH EXPERIENCE

Rensselaer Polytechnic Institute, *Research Assistant (IBM-Funded Collaboration)*, Troy, NY

July 2025 - Present

- Refining a hybrid algorithm for the spin-boson model on both quantum and classical computers.
- Developing analytical solutions for the spin-boson model by performing second-order perturbative calculations of energy eigenvalues and spectral density.
- Benchmarking quantum computers by developing precise analytical solutions used to cross-check a hybrid quantum-classical algorithm (SQD) for simulating open quantum systems.

Fermilab US CMS, *Summer Research Assistant*, Batavia, IL

May 2024 - Aug 2024

- Streamlined interaction with CMS Open Data, enabling deeper analysis of nearly 100 high-energy physics datasets.
- Developed benchmarks for systematic analysis of high-energy physics data using coffea-casa (distributed computing framework) at University of Nebraska-Lincoln.

St. Olaf College, *Quantum Algorithms Student Researcher*, Northfield, MN

Feb 2023 - May 2023

- Implemented Quantum Imaginary Time Evolution (QITE) in Python to compute Hamiltonian ground states.
- Utilized Qiskit to learn quantum circuit fundamentals and simulated components of the QITE algorithm.
- Co-authored an introductory quantum computing textbook for underserved high school students.

St. Olaf College, *CURI Summer Researcher*, Northfield, MN

May 2022 - Aug 2023

- Collaborated in a microscale study with Penn State University to demystify the conditions needed to reach superlubricity of Diamond-Like Carbon where we reduced the coefficient of friction by a factor of 10.
- Presented a poster of our findings to experts in the field alongside our professor at the Tribology Gordon Research Conference 2022 which furthered my knowledge in Tribology.

FUNDING

- IBM-Funded Research Assistantship, Rensselaer Polytechnic Institute, PI: Joel Giedt, (2025–Present).

PUBLICATIONS

- S. Jang, A. G. Colliton, H. S. Flaih, E. M. K. Irgens, L. J. Kramarczuk, G. D. Rauber, J. Vickers, A. L. Ogrinc, Z. Zhang, Z. Gong, Z. Chen, B. P. Borovsky, S. H. Kim, Why is Superlubricity of Diamond-Like Carbon Rare at Nanoscale? *Small*, 2024

POSTERS AND PRESENTATIONS

- **Poster:** "A Microscale study of Diamond-Like Carbon superlubricity," Tribology Gordon Research Conference, 2022
- **Poster:** "Microscale study of Diamond-Like Carbon superlubricity," Carleton College, 2023
- **Poster:** "Implementing CMS Open Data from 2016 into the AGC Coffea workbook", Science Communication Conference, St. Olaf College, 2025
- **Poster:** "Benchmarking the Spin-Boson Model on Quantum Computers", APS Denver Conference, 2026

TEACHING EXPERIENCE

St. Olaf College, Teaching Assistant & Grader (Partial Differential Equations), Northfield, MN Sept 2024 - Dec 2024

- Selected by faculty to grade and mentor students in Partial Differential Equations.
- Provided curated feedback on mathematical logic for 30 students weekly curated to their level.

St. Olaf College, Teaching Assistant & Grader (Linear Algebra), Northfield, MN Feb 2022 - May 2022

- Selected by faculty to grade and mentor students in Linear Algebra.
- Provided detailed feedback on mathematical proofs and logic for 26 students weekly, shifting the focus to conceptual understanding.

LEADERSHIP EXPERIENCE

St. Olaf College, Physics TEAM-UP Together Associate, Northfield, MN Sept 2021 - May 2025

- Collaborated to build an inclusive physics community, including the creation of the "PhLounge" for 120+ students.
- Reformed the Physics department mission statement based on faculty and student body feedback.
- Coordinated outreach experiences that demonstrated science demos

St. Olaf College, TEDx Finance and Logistics Coordinator, Northfield, MN Oct 2022 - Apr 2023

- Managed a budget and sourced TEDx Letters, carpet and decor for the event.
- Coordinated event flow for an audience of 100+ attendees

SKILLS

- *Computer Languages:* Python (Advanced), R (Advanced), Mathematica (Advanced), SQL, LaTeX, Labview
- *Scientific Computing:* Qiskit, Numpy, Matplotlib, Pandas, Scikit-learn
- *Research Methods:* Monte Carlo Simulations, Hamiltonian Dynamics, Perturbation Theory, Machine Learning
- *Tools:* Git/Github, Unix, High-Performance Computing, Fusion360
- *Language Skills:* English, Jamaican Creole, Spanish